
ALZHEIMERNET: MULTIMODAL ALZHEIMER DISEASE CLASSIFIER WITH DUAL EXPLAINABILITY

Manuel Carlucci

Department of Computer Science
University of Bari Aldo Moro
Bari, Italy
m.carlucci69@studenti.uniba.it
Mtr: 855237

Vittorio Nitti

Department of Computer Science
University of Bari Aldo Moro
Bari, Italy
v.nitti19@studenti.uniba.it
Mtr: 856698

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ABSTRACT

Alzheimer’s disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia, marked by irreversible cognitive decline. Mild Cognitive Impairment (MCI) is an intermediate stage that often precedes AD, making early and accurate diagnosis essential for timely intervention. In recent years, machine learning has markedly improved the automated diagnosis of AD from neuroimaging.

However, two challenges limit clinical adoption: most high-performing models operate as opaque "black boxes," offering no insight into why a prediction is made, and distinguishing cognitively normal (CN) subjects from MCI remains difficult due to subtle anatomical differences and high inter-subject variability.

To address this, we propose **AlzheimerNet**, an interpretable framework that couples a **2.5D ResNet-18 classifier with dual explainability** (Grad-CAM saliency and per-class multimodal reconstruction-based anomaly mapping) to deliver diagnosis that is both accurate and explainable.

The model produces calibrated probabilistic predictions for the multi-class task (CN vs. MCI vs. AD) and for all three binary tasks (CN vs. MCI, MCI vs. AD, CN vs. AD), while the two explainability branches expose the anatomical regions driving each decision, linking model outputs to clinically meaningful evidence.

We evaluate the framework on the ADNI4 dataset, which considers structural MRI (T1, T2, FLAIR). A reconstruction-based classifier trains three independent, class-specific multimodal autoencoders (CN, MCI, AD) and compares each subject’s scan against its own reconstruction: the per-pixel discrepancy between input and reconstruction is used both to assign the subject to the class whose decoder reconstructs it most faithfully, and to generate a spatial anomaly map that localizes exactly where the anatomy departs from that class’s expected pattern. This classifier reaches a weighted F1 of 0.52 after bias calibration and a post-hoc MCI fallback rule.

The supervised 2.5D classifier substantially improves on this, achieving a balanced accuracy of 0.468 (AUC 0.724) on the three-class task and 0.690 (AUC 0.687) on the challenging CN vs. MCI distinction on the held-out evaluation split, with the CN vs. AD task reaching the strongest separation (balanced accuracy 0.650, AUC 0.828 on the test split).

Beyond predictive performance, the generated Grad-CAM and anomaly-map explanations consistently concentrate on neural structures in line with known AD atrophy patterns (e.g. hippocampus area), supporting clinical trustworthiness even in misclassified cases.

These results indicate that interpretability and diagnostic accuracy can be pursued jointly, offering a transparent decision-support tool for early AD detection, while also revealing that MCI’s transitional neuroanatomical profile remains a fundamental obstacle to reliable three-way staging.

1 Introduction

Alzheimer’s Disease (AD) is a progressive neurodegenerative disorder that develops over many years and is typically preceded by Mild Cognitive Impairment (MCI), a prodromal stage characterized by measurable cognitive decline with largely preserved daily functioning. Since many MCI cases progress to AD, accurately distinguishing cognitively normal (CN), MCI, and AD subjects is clinically crucial for early diagnosis, prognosis, and intervention. However, this task is challenging due to the subtle and highly variable brain structural changes that characterize early disease stages, particularly between CN and MCI. Key biomarkers such as hippocampal atrophy [17], cortical thinning, and white-matter alterations are informative but distributed across the brain and insufficient on their own to reliably separate adjacent stages, making the problem well suited to data-driven approaches. Deep learning, particularly convolutional neural networks, has shown strong potential in learning discriminative imaging patterns directly from structural MRI without relying on manual feature engineering, outperforming traditional pipelines on datasets such as ADNI. Motivated by this, the goal of this work is to develop a deep learning system that classifies subjects along the AD continuum (CN, MCI, AD) using structural MRI, formulated both as a multi-class problem and as three binary discrimination tasks. The model is designed to produce probabilistic outputs for each class and to operate at the patient level, integrating information across complementary MRI modalities and multiple views to reflect clinical diagnostic practice rather than relying on isolated images.

Beyond classification performance, a central requirement is interpretability. The system is explicitly designed to highlight the brain regions that most influence each prediction, enabling clinicians to assess whether decisions align with known neuropathological patterns rather than spurious correlations such as imaging artefacts or scanner-specific effects. This emphasis on explainability is essential in a clinical context, where model transparency and the ability to scrutinize and validate reasoning are necessary for trust, safety, and responsible deployment, especially given the high stakes associated with diagnostic decisions and early intervention strategies. To reinforce this interpretability objective, the classifier is complemented by a second, reconstruction-based strategy built on autoencoders. The idea is to learn, separately for each clinical group, a compact model of what brain imaging characteristic of that group looks like, drawing on all three modalities (T1, T2, and FLAIR) so that their complementary information is combined into a single group-specific representation. When a new scan is examined, the reconstruction produced by each group’s model reveals how well the scan conforms to that group’s learned pattern, and the regions that are poorly reconstructed, the anomalies, highlight where the subject’s anatomy departs from the expected norm. This yields spatial anomaly maps that localize disease-related deviations directly on the image, offering a view of the pathology that is independent of, and complementary to, the saliency maps derived from the classifier. Together, the two approaches provide converging forms of visual evidence, strengthening the anatomical interpretability that is essential in the medical domain. The proposed system is developed and assessed on data drawn from the ADNI-4 cohort, the most recent phase of the Alzheimer’s Disease Neuroimaging Initiative, a large multi-site effort that has become a standard reference for the study of imaging biomarkers along the Alzheimer’s continuum.

2 Related Work

Deep learning approaches to Alzheimer’s disease (AD) detection from structural MRI have been dominated by convolutional architectures, valued for their inductive spatial bias and compatibility with transfer learning from large natural-image datasets, particularly important given the limited size of typical AD cohorts such as ADNI [1].

CNN-based classification. Convolutional networks have been extensively benchmarked across AD datasets and consistently identify *hippocampal and medial temporal lobe atrophy* as the most discriminative regions [2]. More recent hybrid designs combine a ResNet or VGG backbone with Vision Transformer (ViT) self-attention to jointly capture local structural features and long-range connectivity patterns [3]. These architectures achieve state-of-the-art accuracy on curated benchmarks, but typically *operate on a single MRI modality* (T1-weighted) and rely on large or pre-trained models that are impractical under constrained compute budgets.

Graph Neural Networks. A second direction models the brain as a graph whose *nodes correspond to anatomical regions* of interest derived from a brain atlas and whose edges encode structural or functional relationships; Graph Neural Networks (GNNs) are then applied for classification [4]. This paradigm offers anatomically interpretable representations, but its performance is critically *sensitive to the choice of graph topology* and atlas parcellation—a design decision that requires neuroimaging expertise not always available within a team [5].

Multimodal MRI fusion. Combining complementary MRI sequences (T1, T2, FLAIR) consistently improves diagnostic performance over any single modality. Existing fusion strategies range from early concatenation of input channels to late fusion of modality-specific features and attention-based weighting [6, 7]. Despite this evidence, most

published architectures fuse at most two modalities and apply a single, shared encoder to all of them, *discarding any potential for modality-specific or class-specific latent specialisation*.

Autoencoder-based anomaly detection. Reconstruction-based unsupervised anomaly detection trains a model on healthy anatomy and flags deviations in pathological scans as high reconstruction error [8]. UNet variants, which introduce skip connections to preserve high-frequency spatial detail during reconstruction, produce sharper and more localised error maps than vanilla autoencoders [9]. Applications to AD, however, remain inherently binary (healthy vs. anomalous): a single autoencoder trained on cognitively normal (CN) subjects cannot distinguish Mild Cognitive Impairment (MCI) from AD, since both classes deviate from the CN prototype by construction.

Class-specific autoencoders. The idea of training one autoencoder per class and classifying an unseen sample by assigning it to the class whose reconstructor achieves the lowest error has been formalised in the computer-vision literature. Bojanowski and Joulin [10] apply a stack of discriminative class-specific autoencoders to multiclass anomaly detection in medical endoscopy images, using a combined reconstruction, compactness, and inter-class distance loss. *To our knowledge, no prior work applies this paradigm to multimodal neuroimaging for ternary AD staging or combines it with a parallel classification branch and dual explainability signals.*

Explainability in neuroimaging. Grad-CAM and its variants remain the dominant explainability tool for CNN-based models, producing spatial heatmaps that attribute predictions to image regions [11]. For graph-based models, GNNExplainer serves an analogous role by identifying influential subgraphs [5]. Recent surveys on explainable AI in dementia emphasise that interpretability methods are most clinically valuable when *multiple independent explanation signals can be cross-validated against one another*, rather than relying on a single tool in isolation [5]. *The present work follows this recommendation by combining reconstruction-based anomaly maps with Grad-CAM heatmaps derived from the same shared encoder.*

3 Materials

The experiments were conducted on the ADNI4 dataset (Figure 1), comprising structural brain MRI from 817 subjects. Each subject underwent one or more scanning sessions, resulting in 952 sessions in total. In particular, 122 subjects were scanned more than once, at screening, baseline, 12-month, or 24-month timepoints. Every session produced three co-registered volumes: a T1-weighted (MPRAGE), a T2-weighted, and a FLAIR scan, giving 2,856 volumes overall (952 per modality). For each session, the dataset further provides the associated demographic and clinical metadata, namely the subject’s age and sex, the diagnostic label, and the visit timepoint.

Based on clinical assessment, subjects were divided into three diagnostic groups: 466 cognitively normal (CN), 295 with mild cognitive impairment (MCI), and 56 with Alzheimer’s disease (AD), corresponding to 490, 389, and 73 sessions respectively.

The dataset is provided with an official partition into training, evaluation, and test subsets, comprising 664 (69.7%), 141 (14.8%), and 147 (15.4%) sessions, respectively. In the present study, only the official training and evaluation subsets were used for model training and validation, whereas the official test subset was not considered.

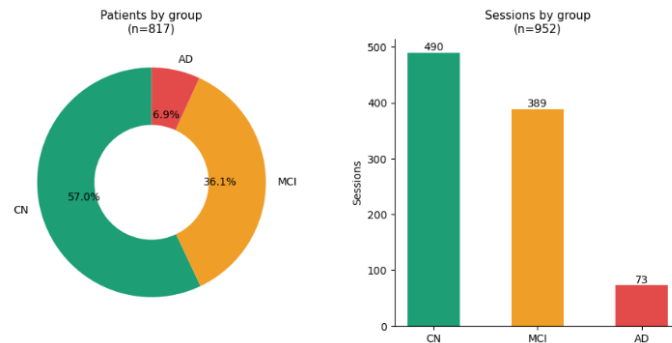


Figure 1: ADNI4 dataset

Prior to training, all MRI volumes underwent a standardized preprocessing pipeline designed to improve data consistency while preserving anatomical information. First, negative voxel intensities were removed through **zero-clipping**, as such values do not represent meaningful tissue intensities. Formally, each voxel intensity was transformed according to

$$I_{\text{clip}}(x, y, z) = \max(I(x, y, z), 0), \quad (1)$$

where $I(x, y, z)$ denotes the original voxel intensity at spatial location (x, y, z) , and $I_{\text{clip}}(x, y, z)$ is the resulting clipped intensity.

Each volume was then **cropped along the axial (z) axis** to retain only the anatomically informative region of the brain. For each axial slice, the amount of brain tissue was estimated by counting voxels above an intensity threshold, and slices whose tissue area fell below a fraction of the maximum observed area were discarded. This procedure removes uninformative slices located near the skull base, as well as near the vertex of the skull, while preserving the central region containing structures such as the hippocampus and lateral ventricles, which are particularly relevant for Alzheimer’s-related classification. Cropping was restricted to the z-axis only, leaving the x and y axes unmodified, in order to avoid introducing spatial bias or excessive geometric distortion on multiple axes.

The cropped **volumes were then resampled** to a fixed spatial resolution of $128 \times 128 \times 128$ voxels using linear interpolation, ensuring uniform input dimensions across all subjects and modalities.

Following resampling, each volume was independently **normalized** to the $[0, 1]$. This normalization reduces inter-subject intensity variability and improves numerical stability during network training.

To enhance local anatomical details, **Contrast Limited Adaptive Histogram Equalization (CLAHE)** was subsequently applied slice-by-slice along the axial direction. A brain mask was generated for each slice to separate brain tissue from the background, and CLAHE was applied only to slices containing a sufficient amount of brain tissue. By operating on local image regions and limiting histogram amplification through a clipping mechanism, CLAHE increases local contrast while reducing the risk of excessive noise enhancement.

For training, a fixed number of axial slices (16 per volume) was sampled from each resampled volume, rather than using the full set of 128 slices. Slices were selected at evenly spaced intervals starting from a fixed axial offset, determined empirically by visual inspection as the point where cerebral tissue first appears, below which the volume contains only skull and bone structures. This choice ensures that no structural brain information is discarded by an overly conservative offset, while excluding clearly non-cerebral regions.

Due to natural inter-subject anatomical variability, the exact tissue content at this offset is not perfectly aligned across patients; a small amount of residual cerebellar tissue may therefore still be present in the lowest sampled slices for some subjects. This is an accepted limitation of the fixed-index sampling strategy, affecting only a minority of the sampled slices, and represents a deliberate trade-off against the added complexity of anatomical atlas-based alignment.

To reduce class imbalance, **online data augmentation** was applied only to the AD class during training. Each original AD volume generated three additional augmented samples, resulting in four training instances per subject. Augmentations were performed on-the-fly and consisted of random left-right flips and small rotations within $\pm 5^\circ$.

4 Methods

4.1 Design Rationale

This section motivates each architectural choice, starting from the task constraints and progressively narrowing down to the final design.

Task definition. The task requires classifying brain MRI scans into three clinical stages – Cognitively Normal (CN), Mild Cognitive Impairment (MCI), and Alzheimer’s Disease (AD) – producing softmax probabilities over the three classes, together with three derived binary tasks obtained by filtering pairs of classes. The model must provide mandatory explainability, and must operate on three MRI modalities (T1, T2, FLAIR).

Why diffusion models were discarded. Diffusion models are generative models whose purpose is to synthesize new images from noise. The task at hand is classification: an existing image is given and a class label must be predicted, with no generative component required. Diffusion models therefore play no role in a pure classification pipeline and were discarded for incompatibility of purpose.

Why a vanilla autoencoder is insufficient on its own. An initial idea was to train an autoencoder exclusively on healthy (CN) images, exploiting the anomaly-detection principle: the model learns the manifold of healthy images, and when presented with a pathological image it reconstructs anomalous regions poorly, considering an error map $|\text{input} - \text{reconstruction}|$ that localizes the anomaly. This idea is conceptually sound but insufficient on its own: a single autoencoder trained only on CN produces a scalar “how anomalous” score, which is naturally binary (healthy vs. anomalous) and cannot separate the two pathological stages, since both MCI and AD would appear equally anomalous relative to the CN prototype. As the challenge requires probabilities over all three classes, this initial single-autoencoder design does not meet the task requirements on its own and motivated the per-class extension described below.

Why a UNet replaces the vanilla autoencoder. The autoencoder was retained as the core reconstruction component but its internal architecture was improved. A vanilla autoencoder compresses all information into a bottleneck and reconstructs solely from it, losing fine detail: reconstructions are blurred even in healthy regions, so the error map is noisy everywhere and fails to localize the anomaly. A UNet is essentially an autoencoder with skip connections: each decoder level also receives the corresponding encoder feature maps, preserving high-resolution spatial information.

From a single shared encoder to one autoencoder per class. The architecture initially discussed used a single shared encoder per modality, branching into a UNet decoder (trained only on CN, for the anomaly map) and a classifier head (trained on all classes). Initially we would have trained one encoder per modality (T1, T2, FLAIR) only on CN, freeze the weights and copy them into 3 new encoders for which we’ll continue the training for the other 2 groups (MCI,AD). This design guaranteed that the anomaly map and the Grad-CAM heatmap were computed on the same latent representation, which made their comparison directly meaningful.

Following discussion with the PhD supervisors, the design was revised into *three independent multimodal autoencoders, one per clinical class* (CN, MCI, AD). Each class-specific autoencoder is itself composed of three modality-specific encoders (T1, T2, FLAIR), whose embeddings are aggregated into a single fused embedding for that class, which is then decoded by a single class-specific decoder. This considers nine encoders in total (3 modalities $\times$ 3 classes) and three decoders (one per class), each decoder receiving the aggregated multimodal embedding of its class. Each autoencoder is trained only on samples of its own class, becoming a specialist reconstruction model for that class’s anatomical pattern.

The encoder architecture is represented in Figure 2.

The decoder architecture is represented in Figure 3.

This revision directly resolves the limitation of the single-decoder design: instead of one anomaly score per subject, three independent anomaly maps are obtained per subject, one per class-specialist autoencoder. For a subject whose true class is, for instance, AD, the AD-specialist autoencoder is expected to have low reconstruction error (it has learned that anatomical pattern), the CN-specialist autoencoder is expected to have high reconstruction error (healthy-pattern reconstruction is a poor fit for AD pathology), and the MCI-specialist autoencoder is expected to have an intermediate error (partial similarity to a transitional pattern).

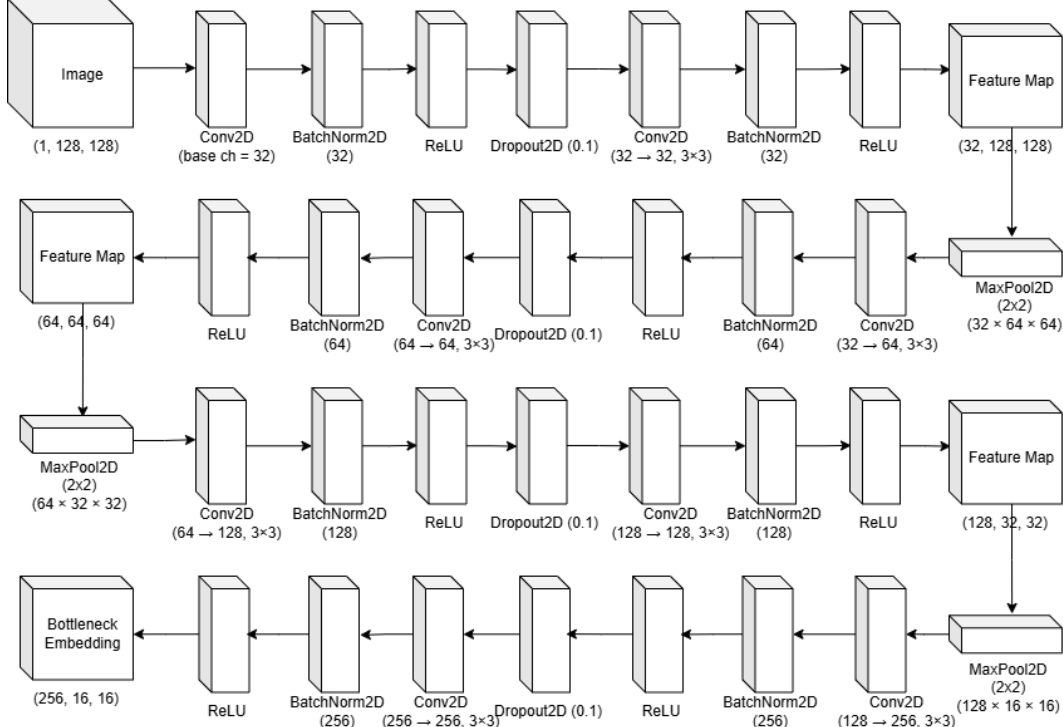


Figure 2: Architecture of the Autoencoders

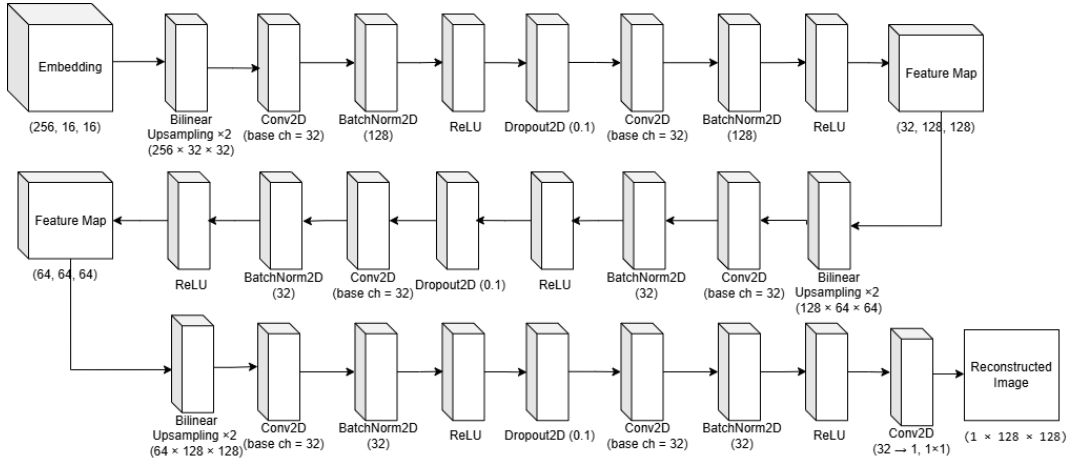


Figure 3: Architecture of the Autoencoders

Why a CNN as the primary classifier backbone. The medical ADNI dataset is relatively small, and *CNNs generalize well even with limited data* thanks to the local inductive bias of convolutional filters. CNNs additionally provide *native compatibility with Grad-CAM*, since they produce 2D spatial feature maps that are the natural target for gradient-based attribution without any architectural adaptation.

Why not a Vision Transformer (ViT). A Vision Transformer *captures long-range relationships better than a CNN* from the first layer, since every patch communicates directly with every other patch via self-attention. This would be useful for Alzheimer’s disease, where pathology is distributed across distant anatomical regions. However, the ViT requires large datasets or heavy pretraining; with a small medical dataset the risk of overfitting is high. In addition, Grad-CAM does not apply directly to ViTs without further adaptation (e.g. attention rollout), which increases implementation cost without a corresponding benefit under the present constraints.

Why not a Graph Neural Network (GNN). A GNN would model the brain as a graph (nodes = anatomical ROIs from an atlas, edges = structural relationships), capturing inter-region relationships with explicit clinical meaning. The main technical concern is that the *graph topology itself is a critical design decision* that must be fixed before training; an inadequate topology would silently cap performance regardless of how well the GNN is trained. Validating or correcting such a choice would require domain expertise in neuroimaging connectomics that was not available within the team.

Why a self-attention module is added after embedding aggregation. A pure CNN builds its global receptive field gradually, layer by layer. For patterns distributed across distant regions, such as Alzheimer’s atrophy (e.g. the hippocampus and temporal cortex, far apart in the image, must be considered jointly), the CNN would need to be very deep to “connect” these regions. A self-attention module inserted after embedding aggregation addresses this by allowing every spatial position to directly weight every other position via query/key/value, independent of spatial distance. This recovers the main advantage of the ViT without its cost: the CNN backbones perform local feature extraction (retaining the benefits of transfer learning and local inductive bias), while attention performs global integration. Since attention operates on 2D spatial feature maps rather than on flattened token sequences as in a ViT, Grad-CAM remains applicable without modification.

Two branches, two fusion strategies: concatenation for reconstruction, projection plus attention for classification. The nine class-modality encoders form a shared trunk read by two branches with different goals, and each branch aggregates the same encoder outputs differently because the two goals are, in a precise sense, opposite.

1. **Reconstruction Branch:** Its goal is to faithfully reconstruct all three modalities (T1, T2, FLAIR), so information from every modality must remain fully preserved and identifiable. For this reason, encoder outputs are fused through **channel-wise concatenation**, a *lossless operation* that keeps modality features in fixed positions. This allows the decoder to reliably recover modality-specific information and generate accurate reconstructions.

During inference, a subject is processed by all three class-specific autoencoders, producing reconstruction-error (anomaly) maps. The autoencoder with the lowest reconstruction error indicates the class whose anatomical pattern best matches the subject. This branch does not directly output class labels; instead, it provides reconstructions, anomaly maps, and an auxiliary class signal.

2. **Classification Branch:** Its goal is not reconstruction but prediction, so it only needs features that are *discriminative for classification*. Since the relevance of T1, T2, and FLAIR may vary across subjects, this branch uses **projection-based attention fusion**.

First, for each class independently, **modality feature maps** are globally pooled, **projected into a shared embedding space**, and combined through **attention pooling**, producing a single class embedding and modality-attention weights that indicate the relative importance of each modality. Repeating this process for all classes yields three class embeddings per subject.

In a second aggregation stage, the three class embeddings are fused into a **patient-level representation**. An optional self-attention layer allows interactions between class embeddings before attention pooling combines them into a final patient vector. The resulting class-attention weights provide an additional interpretability signal, showing which class-view contributed most to the final prediction.

Slice-based processing and the need for per-session aggregation. Each 3D MRI volume is processed as a stack of independent 2D axial slices rather than as a single 3D volume: every encoder consumes one slice at a time, and the corresponding embedding, reconstruction, and reconstruction error are computed per slice rather than per session. In order to understand how the original image is transformed into slices, you can observe the representation below:

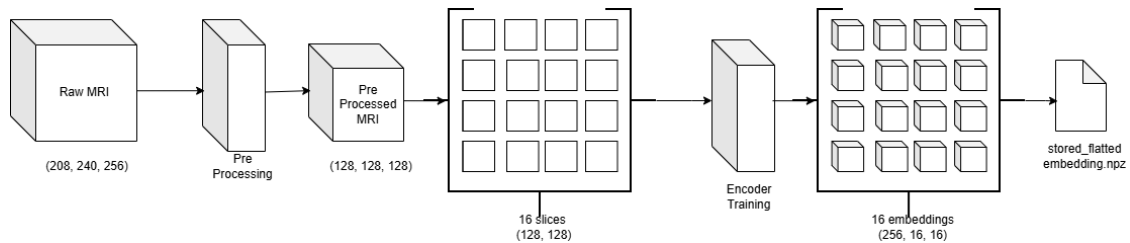


Figure 4: From Image to Slices

Since the challenge requires one prediction per `session_id` rather than per slice, an **explicit aggregation step** is needed to collapse all slice-level embeddings belonging to the same session into a single session-level representation *before any classification head is applied*. Without this step, a classification decision would be based on a single slice, i.e. on a small fraction of the brain volume, which is both statistically unreliable and inconsistent with the submission format.

Slices are not equally informative: peripheral axial slices (near the top and bottom of the skull) contain mostly non-brain tissue or very limited cortical surface, whereas central slices capture the structures most relevant to Alzheimer’s diagnosis, such as the hippocampus and the temporal cortex.

A uniform (unweighted) average across slices would let uninformative peripheral slices dilute the signal coming from diagnostically relevant central slices. To address this, slice-level embeddings are aggregated using a **Gaussian-weighted mean** along the axial axis: slices are assigned a weight following a Gaussian centered on the middle slice of the volume,

$$w_k \propto \exp\left(-\frac{z_k^2}{2\sigma^2}\right), \quad z_k \in [-1, 1],$$

normalized so that $\sum_k w_k = 1$, with σ controlling how sharply the weighting concentrates around the central slices. The resulting weighted-mean feature map is then reduced via **global average pooling** to obtain a single session-level embedding, which is the representation passed to all downstream aggregation and classification steps described below.

Why Grad-CAM is hooked before pooling. Global average pooling, used in the first aggregation stage, discards spatial information by design (appropriate for the classification decision itself, but incompatible with Grad-CAM, which requires a spatial feature map as its target layer). For this reason, the Grad-CAM hook is placed on each encoder’s spatial bottleneck output, upstream of pooling, rather than on any vector produced after it.

The classification tail and the binary heads. The patient vector produced by the second aggregation stage is passed through dropout (active only during training, to discourage reliance on any single feature), followed by a small multilayer perceptron that produces three logits, and a final softmax that converts these into the three-way probabilities

p_{CN}, p_{MCI}, p_{AD} .

Each binary task (CN-vs-MCI, MCI-vs-AD, CN-vs-AD) is handled by a structurally identical head, differing only in that the first aggregation stage is restricted to the two relevant class-embeddings rather than all three, and the final layer produces two logits with a two-way softmax; each binary head is trained only on samples belonging to its two classes.

Why the anomaly maps remain the primary explainability tool. Grad-CAM, applied to the ternary head and to each binary head, indicates where each classifier looked to produce its decision. The three class-specific anomaly maps indicate, independently of any classifier, how far the subject’s anatomy deviates from each of the three learned class-specific patterns. Because every classification head and every autoencoder branch reuses the same underlying class-specific embeddings, the comparison between anomaly maps and the various Grad-CAM heatmaps remains semantically coherent: any divergence between the two signals reflects a genuine difference between “anatomical deviation from a learned class pattern” and “relevance for a specific classification decision,” rather than an artifact of incompatible feature spaces.

4.2 Final Architecture

The final architecture is organized as follows.

Stage 1: class-specific autoencoders. For each of the three classes (CN, MCI, AD), three modality-specific CNN encoders (T1, T2, FLAIR) produce three embeddings, which are aggregated into a single class embedding. This embedding is decoded by a class-specific UNet decoder, trained only on samples of that class, producing a class-specific reconstruction.

Nine encoders and three decoders are used in total. Comparing each reconstruction to the input yields three independent anomaly maps, one per class, whose reconstruction errors collectively serve as a class-discriminative signal.

The described architecture is represented below:

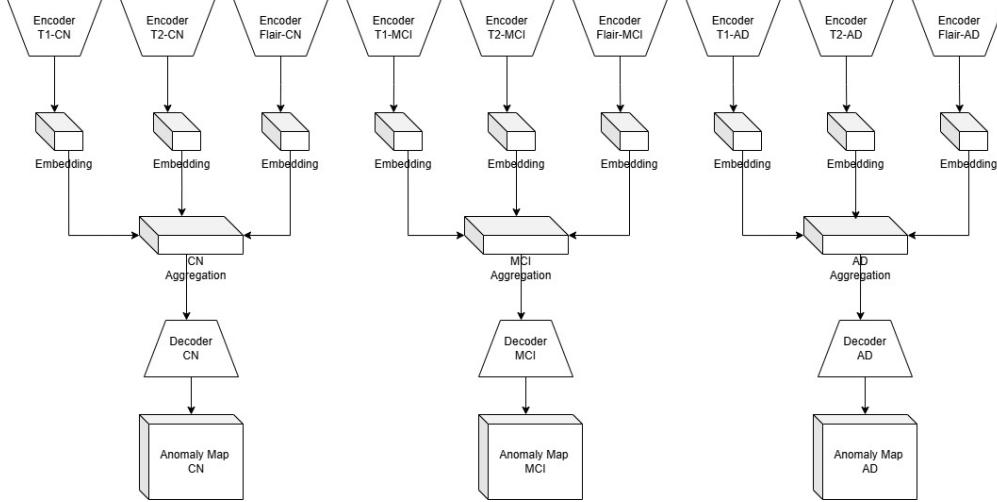


Figure 5: Architecture of the Autoencoders

Explainability. Grad-CAM is hooked on each encoder’s spatial bottleneck, upstream of pooling, and applied to the three-way head and to each of the three binary heads, yielding four Grad-CAM heatmaps in total. These are compared against the three class-specific anomaly maps from Stage 1. The attention weights produced at both aggregation stages (per-modality and per-class-view) are logged as a further, lightweight interpretability signal. Coherence across all signals is guaranteed by the shared use of the class-specific embeddings throughout the architecture.

Stage 2: classification. The same nine encoder outputs are aggregated independently of Stage 1, in two stages. First, within each class, the three modality feature maps are global-average-pooled, projected into a shared embedding space by per-modality linear layers, and combined via attention pooling into one class-embedding; this is repeated for all three classes.

Second, the three class-embeddings are optionally refined via self-attention (allowing each class-view to be re-expressed as a context-aware blend of all three) and then combined, via the same attention-pooling mechanism, into a single patient vector.

For the three-way task, this patient vector is passed through dropout, a multilayer perceptron, and a three-way softmax output (p_{CN}, p_{MCI}, p_{AD}). The architecture can be seen on Figure 6.

For each binary task, the first aggregation stage is restricted to the relevant pair of class-embeddings, and an identical, independently-parameterized pipeline (attention pooling over two tokens, dropout, MLP, two-way softmax) produces the binary prediction. The architecture can be seen on Figure 7.

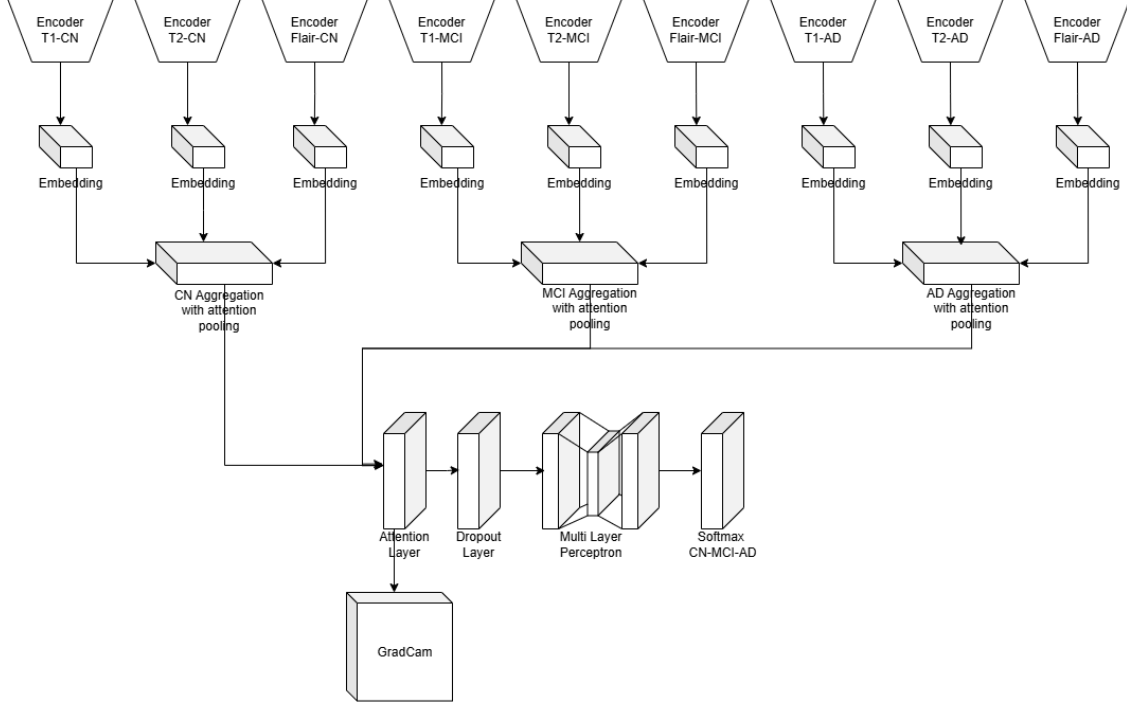


Figure 6: Architecture of the Ternary Classification

Per-group evaluation and bias calibration. For each diagnostic group $g \in \{\text{CN}, \text{MCI}, \text{AD}\}$, the per-slice modality embeddings are concatenated and passed to a group-specific fusion decoder, which reconstructs the multimodal input. The per-slice reconstruction errors ℓ_k are aggregated into a raw session loss via the same Gaussian weights:

$$L_{\text{raw}}^{(g)} = \sum_{k=1}^{16} w_k \ell_k.$$

Since each decoder develops a characteristic reconstruction bias B_g on training data, raw losses are not directly comparable across groups. Each loss is therefore normalized by its expected training bias:

$$L_{\text{calibrated}}^{(g)} = \frac{L_{\text{raw}}^{(g)}}{B_g},$$

and the session is assigned to the group with minimum calibrated loss:

$$\hat{g} = \arg \min_g L_{\text{calibrated}}^{(g)}.$$

A lower calibrated loss indicates that the decoder trained on class g reconstructs the session most faithfully, identifying g as the predicted diagnosis. Section 4.5 details how this calibration and the associated anomaly maps are computed end-to-end at inference time.

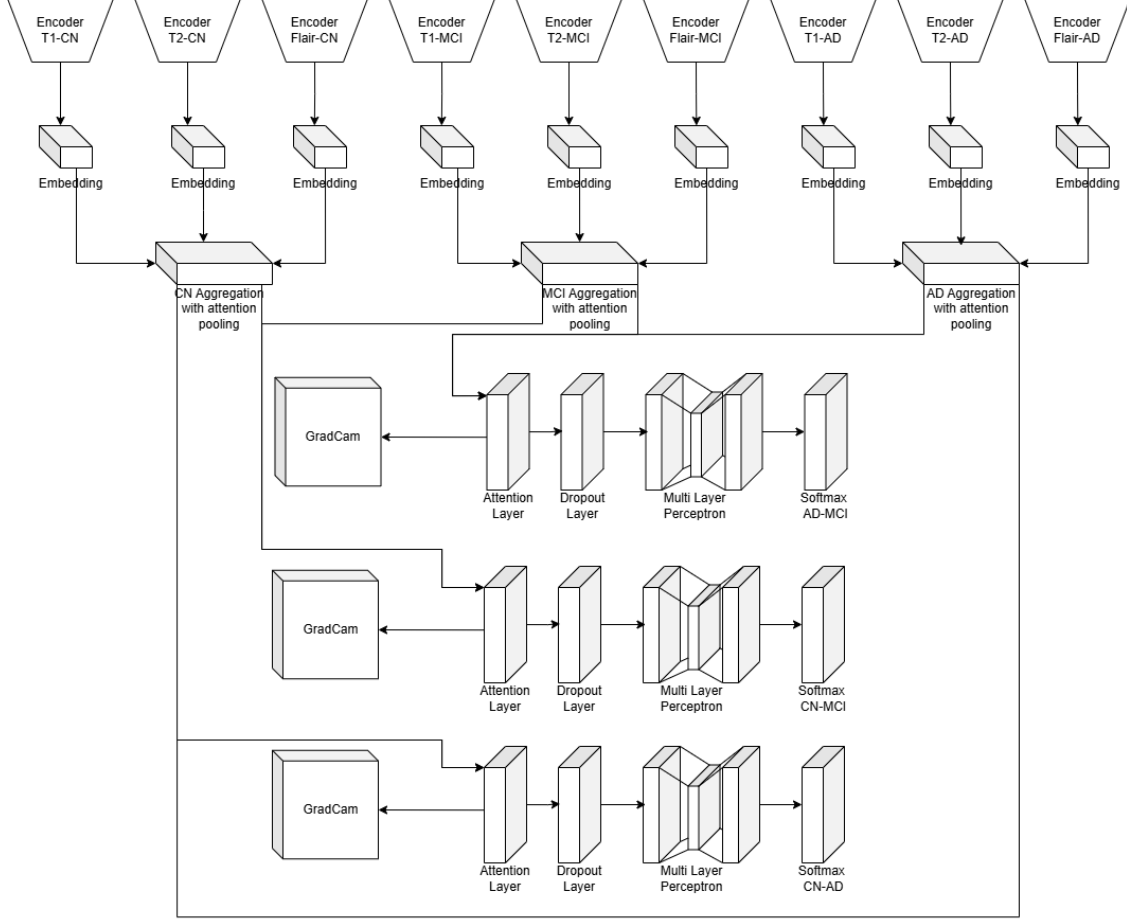


Figure 7: Architecture of the Binary Classification

4.3 Classifier Backbone

Beyond the class-specific autoencoders described above, disease staging itself is ultimately delegated to a supervised convolutional classifier. At the core of this component is a convolutional neural network based on a ResNet18 backbone, designed to predict the disease stage directly from structural magnetic resonance imaging. The network is initialized through transfer learning from large-scale natural image datasets and subsequently adapted to brain MRI, a strategy that is particularly beneficial given the limited size of the available medical cohort. Fine-tuning is performed end-to-end, allowing the pretrained feature extractor to specialize to neuroimaging data while the task-specific classification head is newly defined using a dropout layer followed by a linear layer sized to the number of target classes.

The input representation integrates the complementary information provided by the three MRI modalities, T1, T2, and FLAIR, which are combined into a single multi-channel input. In this way, structural information related to gray-matter atrophy and white-matter abnormalities is exploited jointly, rather than being processed in isolation, enabling the network to learn cross-modal interactions in a manner conceptually consistent with multi-sequence clinical assessment.

Given the volumetric nature of MRI data, each scan is represented through a fixed set of representative axial slices rather than being processed in full 3D, which reduces computational complexity while concentrating training on the anatomical range most relevant to the disease. Because all slices of a subject share the same diagnostic label, the network is trained at the slice level, but prediction is ultimately required at the level of the patient, which is the unit of actual clinical interest. This is achieved through a simple probabilistic aggregation: for every slice, the network outputs a softmax distribution over the candidate classes, and these per-slice distributions are averaged across all slices belonging to the same subject, yielding a single set of class probabilities for that patient. The predicted class is then the one with the highest averaged probability, so that no individual slice is allowed to dominate the decision and the final prediction instead reflects the consensus of evidence gathered across the whole scan. This same aggregation rule is applied consistently at every stage where a decision is required, from selecting the best checkpoint during training

to reporting the final results, so that the model is optimized and evaluated according to the criterion by which it will ultimately be judged.

The classification problem is formulated across four related tasks: the full three-class discrimination among cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer’s disease (AD), together with the three corresponding binary comparisons (CN vs. MCI, MCI vs. AD, and CN vs. AD). To address the strong class imbalance inherent in the dataset, training combines class-weighted loss functions with balanced sampling strategies, ensuring that minority classes contribute meaningfully to gradient updates. Model selection is performed based on patient-level balanced accuracy, prioritizing consistent performance across classes rather than dominance on majority groups.

4.4 Interpretability

Interpretability is treated as an integral requirement of the framework rather than a diagnostic afterthought, and is addressed through two complementary mechanisms operating at different levels of the pipeline: Grad-CAM on the supervised classifier described above, and the reconstruction-based anomaly maps produced by the class-specific autoencoders of Section 4.2.

The first mechanism, Grad-CAM, is applied to the final convolutional layer of the classifier to produce, for each slice, a spatial map of the image regions that most influenced the predicted class. Since a subject-level explanation is preferable to a disconnected set of per-slice maps, these individual saliency maps are combined into a single fused map per patient through a weighted average that favors the more central slices, mirroring the same anatomical relevance criterion used when the representative slices were originally selected. The resulting map can be overlaid on the original scan to indicate, in a form directly interpretable by a clinician, which anatomical regions the classifier’s decision rests upon.

The second mechanism approaches interpretability from a reconstruction-based perspective, reusing the dedicated set of group-specific autoencoders introduced above. For each clinical category (CN, MCI, AD) and each imaging modality (T1, T2, FLAIR), a dedicated encoder compresses axial slices into a compact latent representation, and, within each group, the three modality-specific embeddings are fused and jointly decoded to reconstruct the corresponding multi-modal input. The underlying assumption is that each autoencoder specializes in the anatomical patterns characteristic of its own group, and therefore reconstructs subjects belonging to that group more accurately than subjects belonging to the others. At inference, a subject is passed through all three group-specific autoencoders, and the resulting per-slice reconstruction errors are aggregated across slices, again with greater weight given to central, anatomically informative positions, into a single reconstruction score per group; the subject is assigned to the group with the lowest such score. Rather than discarding the spatial detail of this comparison, the underlying per-pixel squared error is retained and visualized directly as an anomaly map, in which regions reconstructed poorly by a given group’s autoencoder are highlighted as anatomically atypical for that group.

Together, the two interpretability mechanisms offer converging but distinct forms of evidence: Grad-CAM explains a prediction by pointing to the regions that drove the classifier’s decision, while the anomaly maps show, independently of the classifier, where a subject’s anatomy departs from what is expected for a given diagnostic group. Presenting both alongside the model’s output allows its predictions to be corroborated, or challenged, against two independent and anatomically grounded sources of visual evidence. The full inference-time procedure for computing these anomaly maps, together with the successive classifier designs that were evaluated before arriving at the final supervised model, is described next.

4.5 Autoencoder Inference Pipeline

Given a single MRI session comprising three co-registered modalities (T1, T2, FLAIR), the inference pipeline evaluates the session against each of the three group-specific autoencoders (CN, MCI, AD) and produces a calibrated classification together with a pixel-level anomaly map.

Preprocessing. Each of the three modality volumes is independently preprocessed through the same pipeline used at training time: intensity clipping, brain cropping along the axial axis, resampling to a fixed $128 \times 128 \times 128$ resolution, normalization, and slice-wise CLAHE enhancement. This produces three preprocessed volumes of identical shape, one per modality.

Representative slice selection. From the preprocessed volumes, 16 representative axial slices are extracted using the same fixed indices for all three modalities, starting from an empirically determined offset (`start_slice=55`) to exclude non-cerebral structures. A set of Gaussian axial weights is computed over these 16 positions, centered on the middle of the range, so that central slices are assigned higher importance than peripheral ones in the subsequent aggregation step.

Ground truth construction. For each of the 16 selected slices, the corresponding T1, T2, and FLAIR slices are stacked along the channel dimension, producing a $(3, 128, 128)$ target image per slice. Repeating this across all 16 slices yields a target tensor of shape $(16, 3, 128, 128)$, which serves as the reconstruction ground truth for every group evaluated.

Per-group evaluation. Each diagnostic group (CN, MCI, AD) is evaluated independently and identically. For each modality, the 16 slices are encoded slice-by-slice through that group’s modality-specific encoder, producing 16 embeddings of shape $(256, 16, 16)$.

The three modality embeddings are concatenated along the channel axis into a fused representation of shape $(16, 768, 16, 16)$, which is passed through the group’s fusion decoder in a single batched forward pass to reconstruct all 16 slices simultaneously, yielding $(16, 3, 128, 128)$ reconstructions.

The per-slice reconstruction loss is computed as the mean squared error between each reconstructed slice and the corresponding target, averaged over the three channels, producing 16 scalar losses. These are aggregated into a single session-level loss using the previously computed Gaussian weights, and the per-pixel squared error maps are similarly aggregated into a single $(128, 128)$ anomaly map per group.

Bias calibration. Because the three decoders are trained on substantially imbalanced data (49 AD sessions versus 343 CN and 272 MCI sessions), their raw reconstruction losses are not directly comparable: a decoder trained on fewer, less varied samples tends to converge toward a generic reconstruction that yields systematically lower error regardless of the true class of the input. To correct for this, each group’s expected loss is estimated separately on a stratified sample of training sessions, and the raw session loss is divided by this decoder-specific bias before comparison.

MCI fallback. To further improve MCI recall, a **post-hoc MCI fallback rule** was introduced: when the CN and AD calibrated losses for a session are within a tunable margin of each other, the session is classified as MCI, reflecting the clinical definition of MCI as an anatomically intermediate state. A sweep over margin values (0.005–0.100) was performed on the evaluation split; `margin=0.010` was selected as the best trade-off.

Final prediction. The session is assigned to the group whose calibrated loss is the lowest. The corresponding anomaly map highlights, at the pixel level, the regions where the predicted group’s reconstruction diverges most from the original MRI, providing a degree of spatial explainability for the classification outcome.

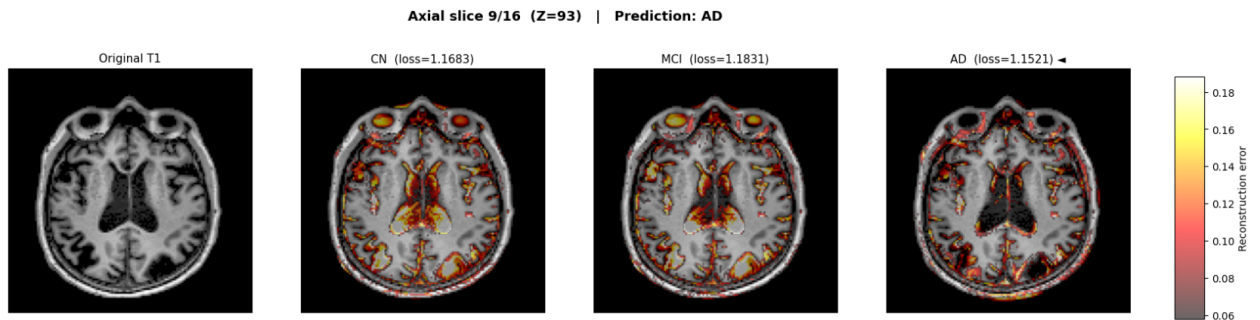


Figure 8: Autoencoder prediction for subject *005_S_10835_sc*, diagnosed with Alzheimer’s disease.

4.6 Multimodal Classification Architecture

Before arriving at the supervised classifier of Section 4.7, an initial classification head was built directly on top of the shared, per-modality autoencoder encoders, once these had been trained and the patient-level embeddings aggregated via Gaussian slice pooling. For each modality $m \in \{T1, T2, FLAIR\}$, the aggregated embedding of shape $(256, 16, 16)$ is first reduced to a 256-dimensional vector through Global Average Pooling (GAP), collapsing the spatial dimensions while preserving channel-wise information. Each pooled vector is then projected into a shared 128-dimensional latent space by a modality-specific linear layer, allowing the network to learn an optimal representation for each modality independently.

The three projected modality vectors are subsequently fused through an attention pooling mechanism: a learned scoring function assigns a scalar importance weight to each modality token, which is then normalized via softmax and used to compute a weighted sum. This produces a single fused patient representation and, as a byproduct, an interpretable set of attention weights indicating the relative contribution of each MRI modality to the final prediction. The fused vector is passed through a dropout layer for regularization and finally through a two-layer Multi-Layer Perceptron (MLP), which outputs class logits converted to probabilities via softmax.

This architecture is instantiated independently for each classification task considered in this work (three-class CN/MCI/AD, and the three binary tasks CN-AD, CN-MCI, MCI-AD): while the modality-wise GAP embeddings are shared across all tasks, the linear projection, attention pooling, dropout, and MLP layers are trained separately for each task, allowing every head to specialize on its specific classification boundary.

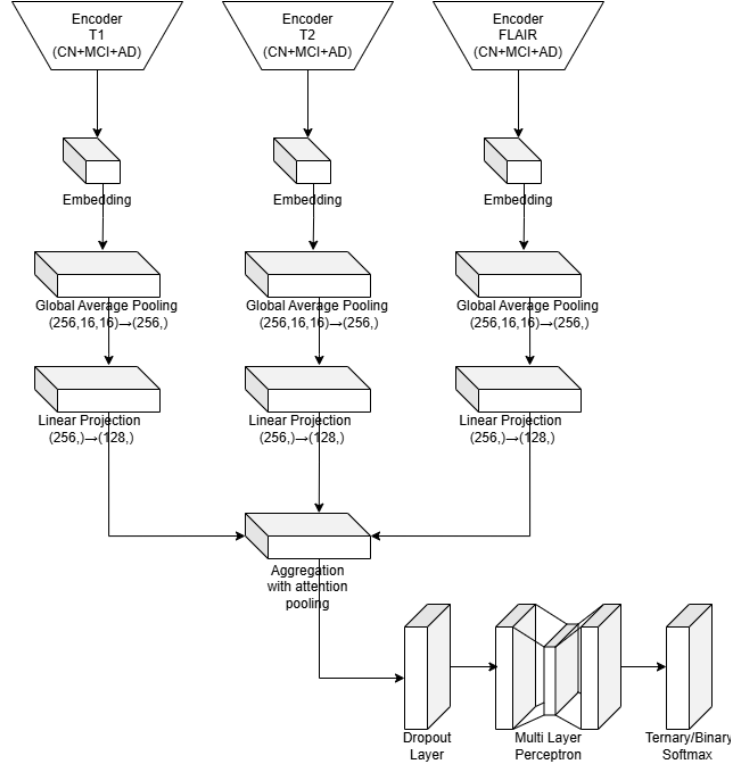


Figure 9: Architecture of the Classification v.1.0

Across all four classification heads, the observed metrics indicate that the classifier is not learning to discriminate between classes, but is instead collapsing to a majority-class predictor. In the three-class CN/MCI/AD task, the training accuracy stabilizes at approximately 0.423, which exactly matches the proportion of CN patients in the training set ($343/811 = 0.423$), while the balanced accuracy converges to 0.333 and the macro F1-score to 0.198. The same degenerate behavior is expected to generalize to the binary tasks, where each head is predicted to always output the majority class of its respective pair. This suggests that the patient-level embeddings produced by the shared, reconstruction-trained encoders lack sufficient discriminative information for the classification task, motivating the shift to a purpose-trained supervised backbone described next.

4.7 2.5D Multimodal Classifier

The reconstruction-loss-based classifier described in Section 4.5, and the attention-fused classifier built on its encoders in the previous subsection, both collapsed to majority-class prediction ($AUC = 0.500$), motivating a redesign around a supervised, transfer-learning approach built independently of the autoencoder embeddings. The resulting pipeline treats each MRI session as a 2.5D volume: rather than processing full 3D scans, a fixed set of $N = 16$ representative axial slices is extracted per modality (T1, T2, FLAIR) using the same Z-cropping and evenly-spaced sampling scheme adopted in the reconstruction pipeline (tissue-based cropping via Otsu-style intensity thresholding, followed by resampling to $128 \times 128 \times 128$, min-max normalization, and slice-wise CLAHE). For each sampled axial position, the three co-registered modality slices are stacked along the channel dimension, producing a 3-channel 128×128 image that is fed directly into an ImageNet-pretrained ResNet-18 backbone, exploiting the natural correspondence between the three MRI modalities and the three RGB input channels.

The classification head replaces the original fully-connected layer with a dropout layer ($p = 0.4$) followed by a linear projection to n classes. Four independent models are trained from the same slice pool: one ternary classifier (CN/MCI/AD) and three binary specialists (CN-MCI, MCI-AD, CN-AD), allowing each decision boundary to be optimized separately rather than relying on a single multi-class softmax.

Class imbalance, which is a persistent issue given the dataset composition (343 CN, 272 MCI, 49 AD training sessions), is addressed at three levels:

- an online `WeightedRandomSampler` that equalizes the per-batch sampling probability across classes;
- inverse-frequency class weights injected into a label-smoothed ($\epsilon = 0.1$) cross-entropy loss
- targeted geometric augmentation of the AD class (random horizontal flip and small-angle rotation, $\pm 5^\circ$, deterministically seeded per session so that all three modalities of a session receive an identical transform), generating up to six training instances per original AD volume

Optimization uses AdamW (weight decay 10^{-4}) with a cosine-annealing learning-rate schedule, mixed-precision (AMP) training, and multi-GPU data parallelism.

At inference time, per-slice softmax probabilities are averaged across all slices belonging to the same session (with augmented copies merged back into their parent session via session-ID normalization) to yield a single patient-level prediction, aligning the evaluation protocol with the clinically meaningful unit of analysis — the patient, not the slice.

The architecture of this classifier is represented in the following figure:

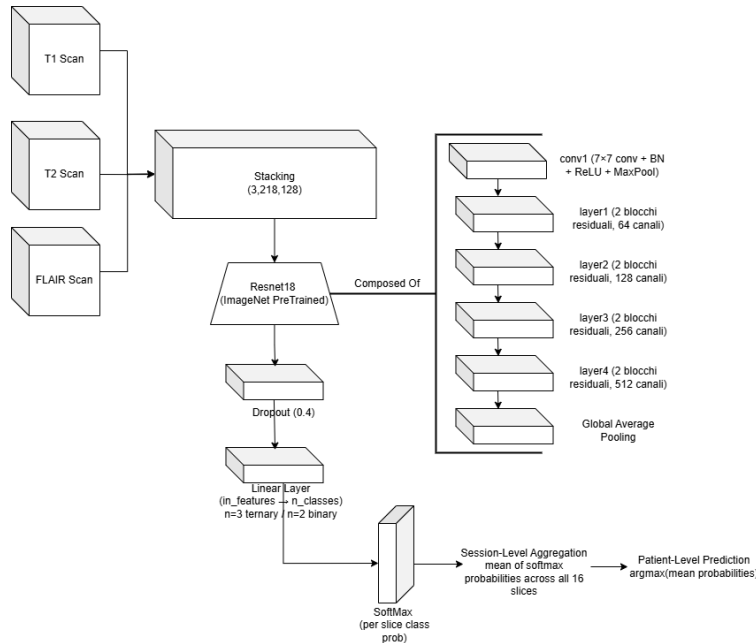


Figure 10: Architecture of the ResNet18 Classifier

4.8 Innovative Aspects

The proposed framework contributes several design choices that, individually and in combination, distinguish it from existing work.

Per-class multimodal autoencoders for ternary staging. Prior reconstruction-based anomaly detection for AD trains a single autoencoder on CN subjects, yielding a binary healthy/anomalous score that cannot separate MCI from AD [8]. We extend this paradigm to three clinical classes by training nine independent encoders ($3 \text{ modalities} \times 3 \text{ classes}$) and three class-specific UNet decoders, each specialised on the anatomical pattern of its own class. At inference, all three autoencoders process the same input; the class whose decoder achieves the lowest Gaussian-weighted reconstruction error across the sampled axial slices is selected as the prediction. This formulation directly resolves the single-class limitation of standard unsupervised anomaly detection, delivering a native three-class probability signal from reconstruction errors alone.

Dual explainability with shared latent space. Because both the reconstruction branch and the classification branch reuse the same nine class-specific encoder outputs, the comparison between the three class-specific anomaly maps (reconstruction branch) and the four Grad-CAM heatmaps (ternary head plus three binary heads) is semantically coherent by construction.

Two-stage multimodal attention fusion. Rather than applying a fixed fusion rule (e.g. concatenation) uniformly across modalities and classes, we adopt a two-level attention pooling strategy.

In the first stage, modality-specific feature maps are projected into a shared embedding space and combined by attention pooling, producing one class embedding per clinical class together with per-modality importance weights.

In the second stage, the three class embeddings are refined via optional self-attention and then pooled into a single patient vector, producing per-class-view importance weights as an additional interpretability signal.

Anatomically-motivated Gaussian-weighted slice aggregation. Inter-slice aggregation strategies in volumetric MRI analysis fall into two main families: non-learnable uniform mean pooling [13], which assigns equal importance to all slices, and learnable soft-attention pooling [14, 15], which infers slice relevance jointly with the classification objective. A third, orthogonal use of Gaussian weighting appears in the literature as a *spatial* intra-slice mask applied to highlight a region of interest within a single slice [16], and is not concerned with inter-slice aggregation.

We propose a distinct, fixed Gaussian prior over the axial slice index:

$$w_k \propto \exp\left(-z_k^2 / 2\sigma^2\right), \quad z_k \in [-1, 1], \quad (2)$$

where w_k is applied as a scalar weight to each per-slice embedding before summation. Unlike attention-based methods, the prior is not learned; it encodes a clinically grounded anatomical assumption that the diagnostically most informative structures for Alzheimer’s disease are concentrated in the central axial slices of the cropped volume, whereas peripheral slices contain predominantly skull base or vertex tissue with limited diagnostic value.

To our knowledge, no prior work applies a fixed anatomically-motivated Gaussian prior along the axial axis as an inter-slice aggregation strategy for brain MRI classification.

5 Results

Autoencoder-Based Classification

The reconstruction-loss classifier, evaluated on 141 held-out sessions (75 CN, 54 MCI, 12 AD), reached a final weighted F1 of 0.52 after progressive refinement through bias calibration and a post-hoc MCI fallback rule (margin = 0.010).

Bias calibration shifts each decoder’s loss by a class-specific offset estimated on a stratified reference sample, correcting for the systematic advantage that imbalanced training confers on minority-class decoders.

The **MCI fallback rule** reassigns a session to MCI whenever the CN and AD calibrated losses are within a tunable margin of each other, reflecting the clinical premise that MCI occupies an anatomically intermediate position between the two. We optimized the margin by reviewing accuracy results and confusion matrix for different margin values. We choose a value of 0.010 as margin because it is a good trade off between the three classes.

Table 1 reports how each refinement step affects accuracy and per-class recall.

Configuration	Accuracy	CN Recall	MCI Recall	AD Recall
No calibration (raw loss)	0.150	0.090	0.040	1.000
Bias calibration (uniform sample)	0.400	0.520	0.240	0.420
Bias calibration (stratified sample)	0.496	0.710	0.220	0.420
Stratified calibration + MCI fallback ($m = 0.010$)	0.511	0.650	0.330	0.420

Table 1: Ablation over calibration strategies. Each row adds one refinement over the previous.

The final per-class performance of the best configuration is reported in Table 2.

Class	Precision	Recall	F1-score	Support
CN	0.670	0.650	0.660	75
MCI	0.450	0.330	0.380	54
AD	0.180	0.420	0.250	12
Weighted avg	0.540	0.510	0.520	141

Table 2: Classification report for the best configuration (stratified calibration + MCI fallback, margin = 0.010).

Loss distribution analysis. Figure 11 shows the MSE loss distributions produced by each group-specific decoder, coloured by true diagnostic group. Across all three decoders the loss ranges overlap substantially (≈ 0.016 – 0.032 MSE), with no decoder producing a clearly separated cluster for its own group.

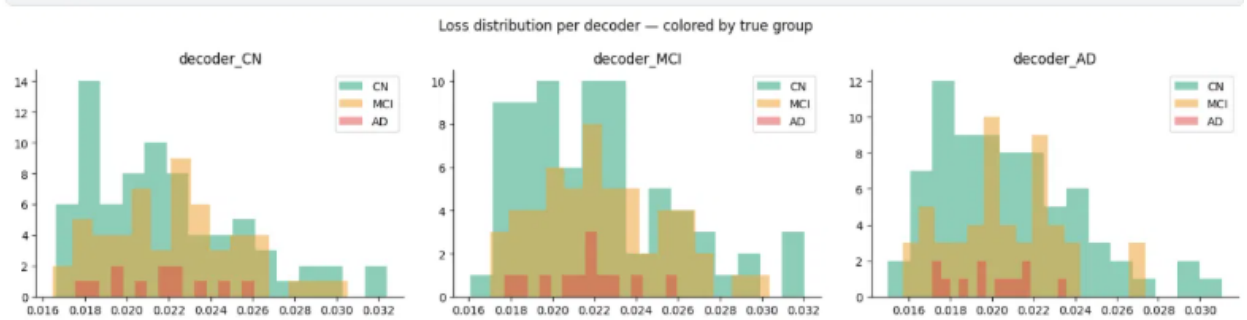


Figure 11: MSE loss distributions per decoder coloured by true diagnostic group (CN, MCI, AD).

Per-slice loss profile. Figure 12 illustrates a representative correctly classified CN session (002_S_6007). The three decoder losses are nearly indistinguishable showing that even a correct prediction rests on a fragile numerical difference rather than a robust discriminative signature.

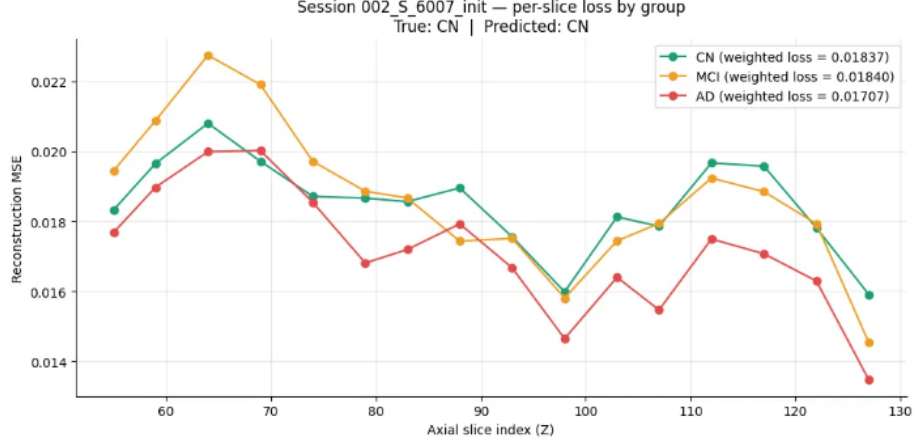


Figure 12: Per-slice reconstruction MSE for session 002_S_6007.

Despite these gains, three structural limitations persist.

- **MCI recall remains low**, consistent with its clinical definition as a transitional state whose neuroanatomical profile overlaps substantially with both CN and AD.
- **AD precision is poor**: with only 12 test sessions the AD results are statistically fragile.
- **reconstruction loss provides a weak discriminative signal**: inter-decoder MSE differences are typically in the range $[0.001, 0.003]$, making the classifier highly sensitive to noise.

These results suggest that a purely reconstruction-based approach is insufficient for reliable Alzheimer’s staging, and motivate the adoption of supervised or contrastive objectives in future work.

5.1 2.5D Multimodal Classifier

The proposed framework was assessed at the patient level for all four classification tasks: the three-class CN/MCI/AD model and the three binary specialist models (CN vs. MCI, MCI vs. AD, and CN vs. AD). For each subject, the posterior probabilities obtained from the sampled axial slices were aggregated through probability averaging to produce a single patient-level prediction. Results are reported for two evaluation settings: the development evaluation split, used for model selection and validation (Table 3), and the final held-out test split, obtained after retraining the selected models on the combined training and evaluation data (Table 4).

Table 3: Patient-level performance of the 2.5D multimodal classifier on the **evaluation split** (train-only models).

Task	Acc.	Bal. Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro Spec.	AUC (OvR)
CN–MCI–AD	0.560	0.468	0.461	0.468	0.463	0.746	0.724
CN–MCI	0.721	0.690	0.730	0.690	0.693	0.690	0.687
MCI–AD	0.545	0.593	0.555	0.593	0.499	0.593	0.552
CN–AD	0.759	0.650	0.597	0.650	0.607	0.650	0.687

Table 4: Patient-level performance on the **test split**, using final models retrained on the combined train+eval pool.

Task	Acc.	Bal. Acc.	Macro Prec.	Macro Rec.	Macro F1	Macro Spec.	AUC (OvR)
CN–MCI–AD	0.578	0.415	0.386	0.415	0.400	0.743	0.666
CN–MCI	0.607	0.591	0.621	0.591	0.574	0.591	0.669
MCI–AD	0.827	0.492	0.419	0.492	0.453	0.492	0.692
CN–AD	0.869	0.542	0.934	0.542	0.541	0.542	0.828

When assessing performance on the **evaluation set**, looking solely at overall accuracy can be misleading due to class imbalance. While the binary CN vs. AD task achieves the highest raw accuracy (0.759), a closer look at the class-balanced metrics reveals that the CN vs. MCI task actually yields the most robust performance. It obtains the highest balanced accuracy (0.690) and macro F1 score (0.693) of all the tasks. This indicates that the model learned

more stable, balanced features for distinguishing healthy controls from early cognitive decline than for the more extreme stages. The MCI vs. AD task proves to be the most challenging binary problem (balanced accuracy of 0.593), reflecting the subtle and heterogeneous structural changes characterizing the later transition into Alzheimer’s disease. As expected, the three-class classification (CN–MCI–AD) represents the most demanding scenario overall, yielding the lowest balanced accuracy (0.468) and highlighting the difficulty of modeling the full disease continuum simultaneously. The limitations highlighted by the balanced metrics are further confirmed when examining the confusion matrices (Figure 13), which reveal the specific misclassification patterns driving these scores. When evaluated on the official **test set**, the models exhibit a similar performance ranking, though with a moderate reduction in metrics. The raw accuracy of the CN vs. AD task jumps artificially high (0.869), alongside a strong AUC (0.828), but the balanced metrics confirm that it struggles with minority class detection (balanced accuracy of 0.542). Once again, the CN vs. MCI classifier proves to be the most balanced and reliable binary model, maintaining the highest balanced accuracy (0.591) and macro F1 score (0.574). For the three-class classification, the transition to the test set reveals a drop in performance, with balanced accuracy decreasing from 0.468 to 0.415, and the macro F1 score dropping from 0.463 to 0.400. Overall, the test-set results confirm a critical insight: relying on raw accuracy heavily skews the interpretation of model success. When class imbalance is properly accounted for through balanced accuracy and F1 scores, the system demonstrates its strongest capability in differentiating cognitively normal subjects from those with Mild Cognitive Impairment.

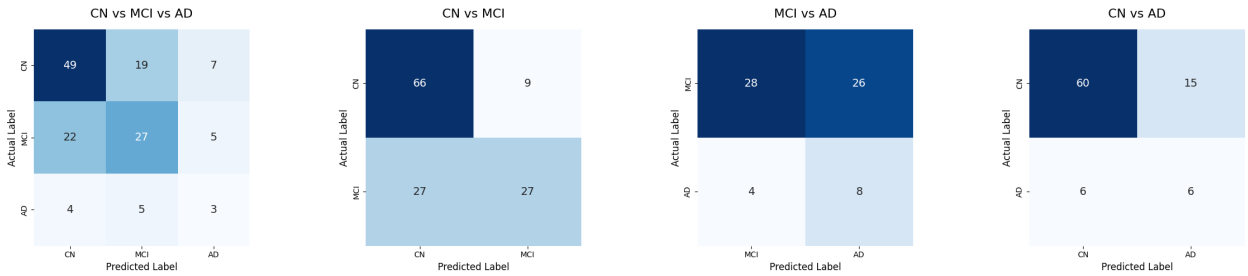


Figure 13: Confusion matrices for each task on the evaluation set

5.2 Grad-CAM Explainability

To assess whether the 2.5D classifier’s decisions are anatomically grounded rather than incidental, Grad-CAM activation maps were computed from the last convolutional block of each ResNet-18 model and fused across the 16 sampled slices using the same **Gaussian axial weighting** scheme employed elsewhere in the pipeline, so that central, **anatomically informative** slices dominate the fused attention map.

A **sanity metric** quantifying the fraction of Grad-CAM *energy falling outside the brain mask* was used to flag spurious activations concentrated on background or skull regions.

For the three case-study sessions discussed above, the fused Grad-CAM overlays for the correctly classified tasks (CN prediction for 002_S_6404_init; MCI prediction for 006_S_4713_init; AD prediction for 381_S_10511_sc under the CN–AD model) consistently concentrate activation around the medial temporal lobe, hippocampal, and periventricular regions rather than on peripheral cortex or background, **in line with the neuroanatomical structures** known to be most affected by Alzheimer’s-related atrophy.

The Grad-CAM analysis of the misclassified case (381_S_10511_sc under the ternary and MCI–AD models, both predicting MCI instead of the true AD) shows attention patterns qualitatively similar to the correctly classified cases, suggesting that the error stems from genuine overlap in the learned feature space between AD and MCI representations rather than from a spatially misdirected model. This reinforces the interpretation that MCI’s transitional neuroanatomical profile poses a fundamental rather than merely technical challenge for this classification approach.

You can observe the Grad-CAM result for the MRI scan "381_S_10511_sc", which belongs to a patient affected by Alzheimer’s disease. This visualization helps us understand which regions of the brain the classifier focuses on during its prediction.

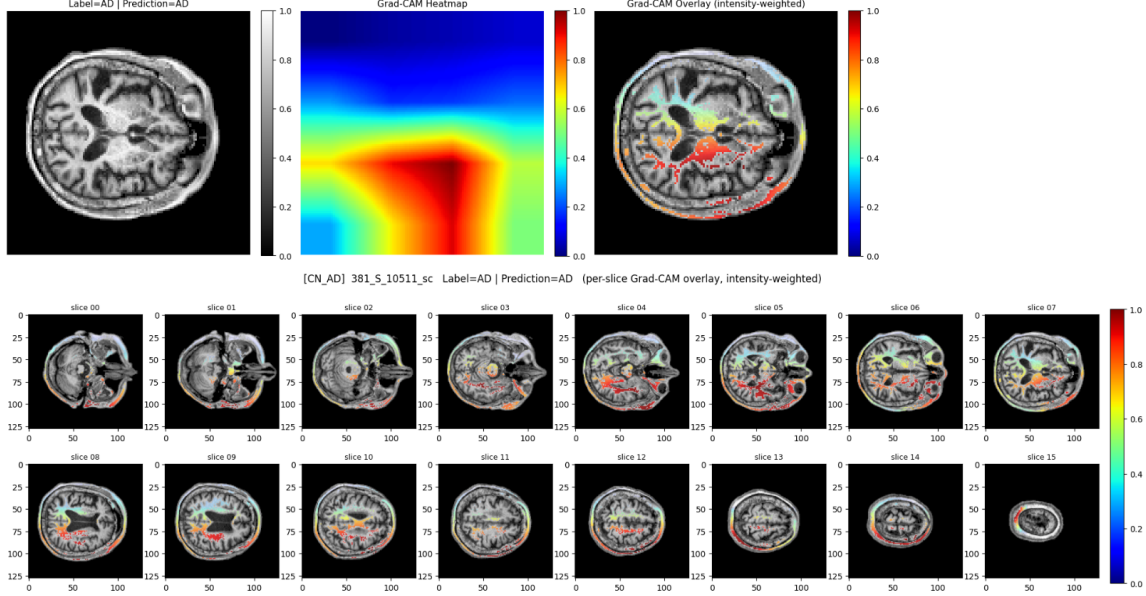


Figure 14: Grad-CAM visualization of MRI scan "381_S_10511_sc" from a patient diagnosed with Alzheimer's disease

6 Conclusion

6.1 Findings

This work developed and compared two complementary approaches to interpretable AD staging on ADNI4.

An initial design (nine class-modality encoders feeding a shared, attention-fused classification head) collapsed to majority-class prediction across all four tasks (three-class training accuracy plateauing at exactly the CN class proportion, 0.423, with balanced accuracy of 0.333 and macro F1 of 0.198), indicating that globally pooled patient-level embeddings from this architecture carried insufficient discriminative signal.

A purely reconstruction-based alternative, using three independent per-class multimodal autoencoders and calibrated reconstruction loss, improved substantially after bias calibration and a post-hoc MCI fallback rule, reaching a weighted F1 of 0.52, but inter-decoder loss differences remained small (0.001–0.003 MSE) and AD precision stayed poor given only 12 test sessions.

Replacing the classification branch with a supervised, transfer-learning 2.5D ResNet-18 classifier trained on stacked T1/T2/FLAIR slices with class-weighted, balanced sampling, yielded the strongest results overall. The CN vs. AD task achieved the best separation (balanced accuracy 0.650, AUC 0.828 on the test split), consistent with the greater anatomical divergence between these two groups. The CN vs. MCI and MCI vs. AD tasks, and the full three-class task, were markedly harder, with MCI recall remaining the weakest link throughout.

Grad-CAM analysis, fused across slices with the same Gaussian axial weighting used for aggregation, confirmed that both correctly and incorrectly classified sessions activate anatomically plausible regions (e.g. hippocampus) rather than background or skull.

Combined with the independently derived reconstruction anomaly maps, which reuse the same class-specific embeddings, this provides two converging, anatomically grounded explanation signals for every prediction.

6.2 Strengths and Limitations

The proposed framework’s principal strength is its dual explainability by construction: because both branches operate on the same class-specific embeddings, any agreement or divergence between Grad-CAM heatmaps and reconstruction anomaly maps is a genuine anatomical signal rather than an artifact of incompatible feature spaces. The 2.5D design also keeps the pipeline computationally tractable on limited hardware while still exploiting large-scale pretraining through the ResNet-18 backbone, and the class-specific autoencoder formulation is, to our knowledge, the first extension of per-class reconstruction-based anomaly detection to ternary AD staging.

Slice coverage. The current pipeline samples only 16 axial slices per volume (out of 128 after resampling) to remain within the memory constraints of Google Colab free tier. This constitutes a significant source of information loss: large portions of the brain volume, including regions known to exhibit AD-related atrophy, may fall between sampled slices and be entirely excluded from the embedding. We attribute part of the gap between our validation performance and published state-of-the-art results to this constraint. Increasing the number of sampled slices to 32 or 64—feasible on machines with larger GPU memory is expected to yield meaningful performance improvements without any architectural change.

6.3 Future Work

Explicit Temporal Progression Modeling The ADNI-4 dataset contains multiple sessions per patient (screening, baseline, 12-month, 24-month), each with its own independent label. A natural extension is to exploit this longitudinal structure by treating each patient’s sessions as a time-ordered sequence and applying an LSTM to the sequence of session-level feature vectors produced by the shared encoder [12].

This is particularly relevant for MCI, which by clinical definition is a transitional stage lacking a stable, uniform anatomical phenotype: subjects labelled MCI may present anatomy anywhere along a continuum between CN and AD.

Patch-based processing. Splitting each MRI scan into a grid of small patches (e.g. 8×8) before feeding them to the nine encoders would, in principle, allow finer-grained, more spatially precise reconstruction and anomaly localization, since each patch could be reconstructed and scored independently rather than relying on a single whole-image embedding per modality. This would also make the anomaly maps more fine-grained, potentially highlighting small, localized anatomical changes that a whole-image reconstruction might average out. However, with nine encoders already active per forward pass in the current design, patch-based processing would multiply the number of forward passes by the number of patches per image, which is not feasible under the computational budget available (Google Colab free tier).

Alternative embedding aggregation strategies. The current Gaussian-weighted mean collapses per-slice embeddings into a single session-level feature map before any classification step. While principled, this aggregation discards all information about within-session spatial ordering. The use of approaches could improve the sensitivity of the session-level representation to localized anatomical features, particularly in the hippocampal region.

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